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Cyberspace Operations

CYBER VULNERABILITY ASSESSMENT/HUNTER (CVA/H) - CYBERCREW TRAINING

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This manual implements Department of the Air Force Policy Directive (DAFPD) 17-2, *Cyber Warfare Operations*. It is consistent with Air Combat Command Instruction (ACCI) 17-202, Volume 1, *Cybercrew Training, and* establishes the minimum AF standards for training and maintaining currency and proficiency for personnel performing crewmember duties on the Cyberspace Vulnerability Assessment/Hunter (CVA/H) weapon system. This publication applies to all Air Combat Command (ACC), AF Reserve (AFR), National Guard Bureau (NGB), and third-party governmental and contract support agencies in accordance with (IAW) appropriate provisions contained in memoranda, support agreements, and AF contracts. This publication does not apply to the United States Space Force or Air Force units executing joint cyber missions under operational control (OPCON) to the Cyber National Mission Force (CNMF). This publication requires the collection and or maintenance of information protected by the Privacy Act of 1974 authorized by Public Law 93-579, as codified at 5 U.S.C. 552a. The applicable System of Record Notice (SORN) F011 AF AFMC B, Patriot Excalibur (PEX) is available at: **http://dpclo.defense.gov/Privacy/SORNs.aspx**. System Records, applies and is available at: **https://dpcld.defense.gov/Privacy/SORNsIndex/DOD-wide-SORN-Article-View/Article/569690/f011-af-afmc-b/**. United States Cyber Command (USCYBERCOM)-defined alternatives to PEX apply, as directed. It is of utmost importance that personnel handling records understand and adhere to the need to take every precaution to protect Personally Identifiable Information (PII) when submitting documents to Department of Defense (DoD) agencies through government internet, software applications, systems, e-mail, postal, faxing, or scanning. This responsibility cannot be overstated. Refer to the following directives for additional guidance: Air Force Instruction (AFI) 33-332, *Air Force Privacy and Civil Liberties Program*,

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SUMMARY OF CHANGES

This publication has been substantially revised and needs to be thoroughly reviewed. Significant changes include but are not limited to Modifying **paragraph 2.4,** and its sub paragraphs, Difference Training to add Type-1 Training Roles and Responsibilities for CVA/H Difference Training along with a CVA/H Type-1 Training Process, step-by-step and updated verbiage and guidance throughout the document to bring ACCMAN 17-2 CVA/HV1 in line with current United States Cyber Command (USCYBERCOM) requirements and guidance for Cyber Protection Teams (CPT). It additionally aligns ACC operational statuses to USCYBERCOM-defined terms. The term "sortie" has been replaced by "event".

Chapter 1—GENERAL GUIDANCE **4**

1.1. General. 4

Table 1.1. ACC and USCYBERCOM Terminology Alignment. 4

1.2. Program Goals. 4

1.3. Roles and Responsibilities. 5

Chapter 2—GUIDANCE AND PROCEDURES **8**

2.1. Qualification Training. 8

2.2. Upgrade Training. 9

2.3. Special Mission Training (SMT). 10

2.4. Difference Training. 10

2.5. Senior Officer Qualification Training. 14
2.6. Squadron Leadership Training. 14
2.7. Requalification Training (RQT). 15
2.8. CVA/H Cybercrew Operational Status. 15
Chapter 3—CONTINUATION TRAINING 17
3.1. Continuation Training (CT). 17
3.2. Currency. 17
3.3. Proficiency. 18
3.4. Proration of Training. 18
Table 3.1. Proration Allowance. 20
3.5. Additional Training. 20
3.6. Currency and Proficiency Reporting. 20
Chapter 4—INSTRUCTOR TRAINING AND QUALIFICATION 22
4.1. Requirements. 22
4.2. Failure to Meet Instructor Requirements. 22
Chapter 5—DOCUMENTATION 24
5.1. Requirements. 24
5.2. Helpful Forms. 24
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 25

Chapter 1

GENERAL GUIDANCE

1.1. General. CNMF directs the use of a jointly procured weapons system that is PURPLE SAFARI (POC: CNMF J6/8/9). Units executing missions under OPCON to the CNMF must comply with training, standardization and evaluation, and weapons and tactics requirements outlined in CNMF guidance; however, unit commanders are encouraged to leverage guidance in this publication to the extent that it is consistent with joint guidance issued by proper authority. This manual prescribes policy and guidance for training CVA/H cybercrew members. Training policy, guidance, and requirements are set forth for all phases of crew training. The objective

1.1.1. Scope. This manual applies to all CVA/H cybercrew members, defined as individuals assigned to utilize the CVA/H weapon system to achieve tasked mission objectives and designated a specific crew position. For this manual, the term "unit" definition includes levels of organizations under Higher Headquarters (HHQ) required to establish a training function. Most units are composed of cyberspace operations squadrons/detachments (hereafter, in this publication, "squadron" will be used synonymously with "detachment"). Additionally, for this manual and uniformity with USCYBERCOM guidance, the term "crew position" will be synonymous with "work role." This manual also applies to members in formal training for immediate assignment to a CVA/H weapon system position. For any publications referenced in this manual, contact ACC/A3/2/6K.

1.1.2. Keywords explained.

1.1.2.1. "Will" or "must" indicates a mandatory requirement.

1.1.2.2. "Should" indicates a preferred, but not mandatory, method of accomplishment.

1.1.2.3. "May" indicates an acceptable or suggested means of accomplishment.

1.1.2.4. "Note" indicates operating procedures, techniques, etc., which are considered essential to emphasize.

1.1.3. Alignment of ACC operational statuses to USCYBERCOM-defined terms. CVA/H is leveraged by U.S. Air Force CPT to execute a USCYBERCOM mission. As such, this volume utilizes USCYBERCOM-defined terminology (such as the proficiency levels of Basic, Senior, and Master) in place of many standard ACC terms, including the use of operational statuses common across other USAF MDS platforms (such as Basic Cyber Qualified (BCQ) and Combat Mission Ready (CMR)). Basic Mission Capable (BMC) status does not have a USCYBERCOM equivalency and will, therefore, remain the same. The most accurate alignment of the ACC operational statuses to USCYBERCOM-defined terms is demonstrated in **Table 1.1** below.

Table 1.1. ACC and USCYBERCOM Terminology Alignment.

ACC	USCYBERCOM
BCQ	Fully Trained (FT)
CMR	Fully Mission Qualified (FMQ)-Basic and FMQ-Senior/Master

1.2. Program Goals. The overall objective of the CVA/H training program is to develop and maintain a high state of readiness for immediate and effective employment across a full range of

cyberspace operations. Mission readiness and effective employment are achieved through developing and mastering core competencies for each crew position aligned to the CVA/H weapon system.

1.3. Roles and Responsibilities.

1.3.1. ACC will:

1.3.1.1. Establish and standardize total force crewmember training requirements in coordination with (ICW) the NGB and Air Force Reserve Command (AFRC) across the Future Years Defense Program (FYDP).

1.3.1.2. Manage all training course requirements and training tasks for the CVA/H weapon system. See AFI 10-217, *Management of Air Force Operational Training and Undergraduate Aircrew Training Systems*, for information on training system management.

1.3.1.3. Chair Training Planning Team (TPT) conferences IAW AFI 10-217. Conference members should include representatives from Squadron Training, Weapons and Tactics, Standardization and Evaluation (Stan/Eval), Operations, Formal Training Unit (FTU), Program Management Office (PMO), and other areas as required.

1.3.1.4. Publish CVA/H Ready Cybercrew Program (RCP) Tasking Memorandum (RTM) annually prior to the start of each fiscal year (FY). The RTM may contain supplemental training requirements or updated guidance.

1.3.2. Numbered Air Forces (NAFs) employing CVA/H will:

1.3.2.1. Maintain oversight of CVA/H cybercrew training within its chain of command.

1.3.2.2. Host conferences and working groups, as necessary, to review training policies and procedures and provide improvement recommendations to lead command for consideration and implementation.

1.3.2.3. Provide approved master training task list (MTTL) item implementation and use guidance ICW HQ ACC.

1.3.2.4. Coordinate on the RTM with leadership, subordinate units, and stakeholders.

1.3.3. Wings or groups employing CVA/H will:

1.3.3.1. Manage the wing's Operations Training program. Wings or groups with an operations support squadrons (OSSs) or similar capability will staff their training office with a Chief of Training and at least one instructor per cybercrew position IAW **paragraph 1.3.5.2.3**. **(T-2)**

1.3.3.2. IAW RTM requirements, forward RTM training reports through HHQ to ACC/A3/2/6K on the 15th day of April and October using the most current guidance provided on the ACC Flight Operations and Training Branch (ACC/A3TO) SharePoint® site. Out-of-cycle reports are authorized should Squadron (SQ)/Commander (CC) (SQ/CC) require HHQ assistance.

1.3.3.3. Assist the Air Reserve Component (ARC) and Air National Guard (ANG) unit training programs as required or requested IAW applicable unit support programs, memorandums of agreement (MOAs), or memorandums of understanding (MOUs).

1.3.3.4. Provide ACC A3/2/6K assistance with the development of work role Upgrade Training plans.

1.3.4. The wing Chief of Training will:

1.3.4.1. Maintain BMC status for at least one CVA/H Weapon System crew position. **(T-2)**

1.3.4.2. Lead training development efforts and standardize training material and programs across the wing or group.

1.3.4.3. Develop and implement programs to ensure training objectives are met.

1.3.4.4. Assist subordinate units in training program management to ensure programs meet mission requirements and provide the necessary support.

1.3.4.5. Establish procedures for review, quality control, and approval of unit training documentation.

1.3.4.6. Review, modify, and implement the RCP as required prior to the start of every FY or upon out-of-cycle RTM publication. **(T-2)**

1.3.4.7. Ensure RCP events are oriented towards maintaining operationally relevant skills and tactical employment. The RTM defines RCP event accomplishment.

1.3.4.8. Coordinate on the RTM with leadership, subordinate units, and stakeholders.

1.3.5. Unit Training Organization:

1.3.5.1. The SQ/CC will:

1.3.5.1.1. Implement the squadron Cybercrew Training Program.

1.3.5.1.2. Manage and implement unit-level training programs. **(T-3)**

1.3.5.1.3. Provide unit training function manpower to execute the duties directed by ACCI 17-202V1 and this manual. **(T-3)**

1.3.5.1.4. Designate cybercrew instructors. **(T-3)**

1.3.5.1.5. Ensure authorized prefixes from the current Department of the Air Force Officer and Enlisted Classification Directories are awarded and affixed to cybercrew members serving in or qualified in Cyber Weapon Systems (e.g. “K” prefix for Instructors, if applicable).

1.3.5.1.6. Ensure cybercrews and individual cybercrew members are provided with adequate post-mission reconstitution time to enable RCP requirements completion.

1.3.5.2. The Training Flight will:

1.3.5.2.1. Be administered and implemented from the unit level.

1.3.5.2.2. Be led by a Chief of Training who reports directly to the Director of Operations (DO).

1.3.5.2.3. Be staffed with a minimum of one qualified instructor for each cybercrew position. All instructors (including the Chief of Training) must maintain FMQ status for their aligned CVA/H cybercrew position. **(T-3)**

1.3.5.2.4. Recommend candidates for instructor upgrade from the most suitable, highest qualified, and most experienced personnel. Requisites for instructor upgrade are outlined in **Chapter 4**.

1.3.5.2.5. Process and maintain training records and forms in a MAJCOM-approved system IAW **Chapter 5** of this document as well as ACCI 17-202, Volume 2, *Cybercrew Standardization/Evaluation Program* and ACCMAN 17-2CVAH, Volume 2, *Cyber Vulnerability Assessment/Hunter (CVA/H) - Standardization/Evaluation Program*. Offline records must also be maintained as a precautionary backup.

1.3.5.2.6. Establish procedures for review and quality control of training documentation. **(T-3)**

1.3.5.2.7. Develop and document an instructor training program designed to train and prepare instructors for qualification. Instructor training programs must be reviewed and approved by the OSS before implementation and must meet the requirements of **Chapter 4**. **(T-3) Note:** For units without an OSS, gain approval through the supporting Group Commander (GP/CC) or their designated representative.

1.3.5.2.8. Provide a quarterly Status of Operations Training brief to unit leadership highlighting Initial Qualification Training (IQT), Mission Qualification Training (MQT), and Continuation Training (CT) status, shortfalls, and missed suspense deadlines.

1.3.6. Cybercrew members will:

1.3.6.1. Complete training, currency, and proficiency requirements within the guidelines of this volume and applicable RTM. **(T-3)**

1.3.6.2. Report additional training gaps or shortfalls for required cyber activities to their supervisor and Chief of Training or training flight representative.

1.3.6.3. Only participate in operational activities they are trained, qualified/certified, current, and approved to conduct, unless under the supervision of a qualified instructor.

1.3.6.4. Ensure the losing organization makes records available to the gaining organization and provides additional hard copies to the member when changing units to maintain documentation of certification, qualification, and training.

Chapter 2

GUIDANCE AND PROCEDURES

2.1. Qualification Training. This section provides CVA/H cybercrew operational status definitions and outlines minimum training requirements for IQT, MQT, Upgrade Training, Special Mission Training (SMT), Difference Training, Senior Officer Qualification Training, and Requalification Training (RQT) for all CVA/H cybercrew positions.

2.1.1. IQT. IQT is defined as formal training required to qualify for basic CVA/H cybercrew duties in an assigned crew position regardless of the unit's operational mission.

2.1.1.1. IQT Prerequisites. CVA/H trainees must complete the USCYBERCOM Cyber Mission Forces (CMF) Formal Training Pipeline required training courses specified for their assigned work role before attending CVA/H formal training. Additionally, trainees must meet the current CVA/H System Training Plan (**https://usaf.dps.mil/sites/ACC-A3/A326K/a326kd/Shared%20Documents/Forms/AllItems.aspx?id=%2Fsites%2FACC%2DA3%2FA326K%2Fa326kd%2FShared%20Documents%2FSTP&p=true&ga=1**) and associated course syllabus identified prerequisites. **(T-2)**

2.1.1.2. Method. The primary method of CVA/H IQT is to attend the FTU course(s) outlined in the CVA/H syllabus and listed in the Education and Training Course Announcement (ETCA) system under the USAF Formal Schools. ETCA can be reached via the Air Force Portal (**https://usaf.dps.mil/teams/app10-etca/sitepages/home.aspx**).

2.1.1.3. In-Unit IQT. When formal course attendance is impractical or quotas are unavailable, units will request waivers to conduct in-unit IQT using formal school courseware. Accomplish in-unit training IAW applicable formal school courseware and the guidance contained in ACCI 17-202V1. **(T-2)**

2.1.1.4. IQT will be taught by qualified instructors as defined in **Chapter 4**.

2.1.1.5. Personnel in IQT will not have supervisory responsibilities or additional duties except as defined within the scope of classroom responsibilities (e.g., class leader, class superintendent, class academic lead).

2.1.1.6. Personnel with previous CVA/H experience (i.e., they were assigned FMQ or BMC status in CVA/H at a unit prior) but were most recently assigned to a non-CVA/H unit (i.e., the unit they transferred from did not use CVA/H and member is no longer current), may be required to re-accomplish the CVA/H crew position portion of IQT. Units will assess the individual's knowledge and may direct the cybercrew member to accomplish all, some, or none of the IQT requirements SQ/CCs may tailor programs for individuals based on previous experience, qualifications, and documented performance.

2.1.2. MQT. MQT is a unit-level, formal training program, developed IAW Instructional Systems Development (ISD) principles that prepares newly assigned cybercrew members to accomplish the unit's mission (see **paragraph 2.7** for RQT). This section prescribes the minimum training requirements to prepare individuals for qualification in their unit's missions and is provided to assist SQ/CCs in developing an MQT program.

2.1.2.1. MQT Pre-requisites. Each cybercrew member must complete all applicable IQT requirements IAW **paragraph 2.1.1** before entering MQT.

2.1.2.2. Time Limit. The cybercrew member will begin in-unit MQT no later than (NLT) 45 days (90 days for the ANG/AFR) of arrival at their gaining unit and upon completion of IQT, security clearance, and indoctrination requirements, whichever occurs last unless the unit commander grants an extension in writing. **(T-2)**

2.1.2.3. Method. The primary method of CVA/H MQT is to attend and complete an in-unit formal training course focused on unit-specific, mission-related training requirements. For all attendees, MQT will take priority over non-training-related duties, including mission execution.

2.1.2.4. MQT will be taught by qualified instructors as defined in **Chapter 4**.

2.1.2.5. Mission-Related Training. Units will develop training addressing areas pertinent to the mission as determined by the SQ/CC. Additional mission-related training elements are addressed in Air Force Tactics, Techniques, and Procedures (AFTTP) 3-1. (U) *CVA/Hunter* (S//NF); AFTTP 3-1.(U) *Advanced Cyber Threat Guide Chapter 13* (S//REL FVEY); AFTTP 3-1. (U) *Integrated Planning and Employment (IPE)* (S//NF); AFTTP 3-3.*CVA/Hunter*; AFTTP 3-3.*IPE;* USCYBERCOM Cyber Warfare Publication (CWP) 3-33.4, *Cyber Protection Team (CPT) Organization, Functions, and Employment*; and other mission-related documents.

2.1.2.5.1. Operations Training Plan (OTP). Units will annually develop, review, and document MQT syllabi and other training requirements in an OTP. **(T-3)** The OTP contains the knowledge, tasks, methods of training, and total training time (to include estimated and maximum training times and lapses between training and events). Schedule, conduct, and document MQT IAW the OTP. While MQT is mission-specific, units will use standardized training whenever possible. Wings or groups may develop a common, foundational OTP that squadrons augment with additional mission-specific training tasks. To develop OTPs, units must align mission objectives and incorporate performance measures, mission essential tasks, local area requirements and procedures, and core functions outlined in USCYBERCOM Cyber Technical Manual (CTM) 7-0.2, *CMF Training and Readiness (T&R) Manual.* Specific Mission Tasks. Specific mission tasks are derived from mission objectives and include other relevant tasks as determined by the unit. **(T-3)**

2.1.2.5.2. Mission objectives and tasks are the minimum requirements for MQT. Additional training events based on gaps in student proficiency or training background, as well as student non-progression, should be made available within the constraints of the training plan and may be added at the SQ/CC's discretion.

2.1.2.6. Certification/Qualification. Upon MQT completion, Initial (INIT) Qualification (QUAL) and Mission (MSN) Evaluations will be conducted IAW ACCMAN 17-2CVAHV2 and certified by the SQ/CC or certifying official (first operational commander in the member's chain of command, or designee).

2.2. Upgrade Training. Cybercrew members pursuing a CVA/H upgrade position will complete Upgrade Training. Upgrade Training is given to operators upgrading from their current crew position to a position of additional responsibility within the CVA/H (e.g., Instructor, FMQ-Basic to FMQ-Senior, HA/NA to Crew Lead). The upgrade position will require further certification or qualification criteria and must contain all requirements of the initial position. Tasks expected for

both positions (e.g., completion of formal training courses) do not need to be retrained if the knowledge and skills are duplicated. CT requirements are only levied against the upgrade position as it is utilized in place of the original position. However, the operator can perform operations for either the original or upgraded position.

2.2.1. The certifying official will determine and assign cybercrew members to train for and maintain upgrade positions.

2.2.2. The certifying official will review and approve all Upgrade training requirements. The Chief of Training will provide appropriate criteria to all candidates for upgrade.

2.2.3. Unless expressly prohibited or restricted by weapon system operating procedures, specific theater Operations Orders (OPORDs), or specific HHQ guidance, upgrade training may be conducted during operational missions with appropriate supervision and IAW instructor requirements found in **Chapter 4**.

2.2.4. All Upgrade Training will be documented IAW **Chapter 5**.

2.3. Special Mission Training (SMT). SMT is given to cybercrew members obtaining a special mission certification on the CVA/H weapon system. SMT can also teach any skill necessary to carry out the unit's assigned mission that every cybercrew member does not require. Specialized training may include special tactics, weapon system capabilities, responsibilities, and operations on specific mission systems/terrain needed to maintain proficiency.

2.3.1. Unless specifically prohibited or restricted by weapon system operating procedures, specific theater OPORDs, or specific HHQ guidance, SMT may be conducted during operational missions with appropriate supervision by a qualified instructor certified in the associated special mission certification.

2.3.2. Special Mission Training Plan (SMTP). Units will develop, review, and document SMT syllabi and requirements annually in an SMTP. The SMTP contains the knowledge, tasks, methods of training, and total training time (to include estimated and maximum training times and lapses between training and events). Schedule, conduct, and document SMT IAW the SMTP. Units will use standardized training whenever possible. Wings or groups may develop a common, foundational SMTP that squadrons augment with additional SMT tasks.

2.3.3. The certifying official will determine and assign cybercrew members that will train for and maintain special mission certifications.

2.3.4. The certifying official will determine and document any CT requirements needed to maintain special mission certifications.

2.3.5. All SMT will be documented IAW **Chapter 5**.

2.4. Difference Training. Difference training will be required when significant modifications or upgrades are made to the CVA/H weapon system. Difference training is typically accomplished after Type-1 training is completed and/or directed by ACC/A3/2/6K. Type-1 training is defined in the CVA/H STP but is commonly referred to as 'Train the Trainers' **(T-3)** focused training. After Type-1 training is complete, difference training will be given to the crew force to provide immediate training and to fill the operator knowledge/performance gap.

2.4.1. ACC A3/2/6K will determine if a modification or upgrade to the CVA/H weapon system is deemed "major". This includes significant hardware, software, or procedural changes that impact unit mission requirements.

2.4.2. Once a significant modification or upgrade has been determined, the CVA/H PMO will develop and deliver Type-1 training along with the full list of Hardware/Software changes, including version numbers for the corresponding OSS (or equivalent organization), supporting wings or groups, and relevant Formal Training Units.

2.4.3. Failure to complete required difference training does not, by itself, disqualify members from maintaining FMQ status. However, untrained members will not perform tasks associated with the new weapon system capabilities until trained unless under the supervision of an instructor/evaluator. **(T-2)**

2.4.4. All Difference Training will be documented IAW **Chapter 5**.

2.4.5. Type-1 Training Roles and Responsibilities for CVA/H Difference Training.

2.4.5.1. MAJCOM (ACC/A3/2/6K) will:

2.4.5.1.1. Serve as the primary training interface between the PMO and assigned cyber operations units.

2.4.5.1.2. Chair the TPT to identify new/ongoing training requirements.

2.4.5.1.3. Advocate for resources for all stages of training and will provide input, as requested, to the MAJCOM Functional Managers (MFMs) regarding initial skills training (IST) as appropriate.

2.4.5.1.4. The MAJCOM, through the Weapon System Team (WST) directs the PMO (Air Force Life Cycle Management Center (AFLCMC)) to:

2.4.5.1.4.1. Execute life-cycle-specific training requirements.

2.4.5.1.4.2. Coordinate with the MAJCOM to receive identification of Type-1 initial cadre for Type-1 training.

2.4.5.1.4.3. Deliver training to Type-1 initial cadre.

2.4.5.1.4.4. Coordinate Type-1 training deliverables to accompany the fielding of new or modification to existing CVA/H capabilities to ensure the Type-1 initial cadre can adequately develop and sustain difference training. Actual deliverables will vary by fielding capability as required, but at a minimum, will include a lesson coversheet, teaching plan, and in-person training session(s) by an expert in the field, recorded and made available for later reference as needed across the force. As for other varying deliverables, a generic list follows:

2.4.5.1.4.4.1. References (e.g., commercial-off-the-shelf (COTS) manuals/products, technical orders (TO), referenced TOs/Time Compliance TOs, commercial terms and definitions list).

2.4.5.1.4.4.2. Product training (e.g., classroom training, computer-based training, commercial training, training credits).

2.4.5.1.4.4.3. Training pre-requisites (e.g., licenses, software, hardware, labs,

scenarios, elevated privileges, access/permissions lists).

2.4.5.1.4.4.4. Task list (e.g., knowledge task, performance task, proficiency level, references for each task).

2.4.5.1.4.4.5. Criterion objectives (addresses the condition (situation under which a student is expected to demonstrate a behavior); behavior (an observable, measurable skill, knowledge, or attitude); and standard (the criteria for acceptable performance by the student)).

2.4.5.1.4.4.6. Lesson plans/courseware.

2.4.5.1.4.4.7. Performance practice scenarios (e.g., labs, exercises).

2.4.5.1.4.4.8. Train Type-1 initial cadre how to teach the materials (e.g., train the trainer, instructor guide, lesson plan review).

2.4.5.1.4.4.9. The associated PMO will provide and sustain the equipment and software, including licensing, required to teach the new capability in units and at the designated FTU.

2.4.5.1.4.5. Coordinate post-training surveys through ACC/A3/2/6K and the 67th Cyberspace Wing Training Office (67 CW/A3T).

2.4.5.2. 67 CW/A3T and Operational Units will:

2.4.5.2.1. Participate in Initial Planning Teams (IPTs) to identify the affected crew positions and establish the appropriate training for each crew position. This will ensure that the PMO delivers adequate Type-1 training.

2.4.5.2.2. Provide pre-vetted names to be designated as Type-1 Initial Cadre for the capability. The operational units and wing training office will nominate appropriate and adequate numbers of personnel with reasonable retainability and do not have outbound assignments. Additionally, at least a portion of the Type-1 initial cadre needs qualified instructors to facilitate the development of MQT and train the remainder of the crew members.

2.4.5.2.3. Coordinate with the PMO to ensure the Type-1 training being developed for the major updates to weapon system releases/features is of significant value. Units are not required to send operators in person to the PMO during the development of Type-1 training; however, there is an expectation that a qualified person will be available to answer/respond to questions/concerns brought forth by the PMO. Training may fall under one or more of the following categories:

2.4.5.2.3.1. Tactical employment training (e.g., difference training) on new or upgraded capabilities or features.

2.4.5.2.3.2. Tactical employment training focused on the “whole of the system” (e.g., Mobile, Deployable, and Garrison Interceptor Platforms (MIP/DIP/GIP)).

2.4.5.2.3.3. Academic fundamental training that explains foundational concepts not covered in the current training pipeline as they pertain to new or updated capabilities or features.

2.4.5.2.4. Ensure all Type-1 initial cadre personnel complete post-training surveys to ensure the training delivered was adequate.

2.4.5.2.5. Determine what additional training, if any, is required beyond the Type-1 Training provided by the PMO.

2.4.5.2.6. Ensure that difference MQT is developed to enable all the crew members to be trained on the new capability organically.

2.4.5.2.7. Work with the Operations Training Development (OTD) team to ensure that the new capability tasks are incorporated into the MTTLs, and when appropriate, nominate IQT updates during the next Syllabus Review Board (SRB).

2.4.5.2.8. Ensure Type-1 initial cadre or other qualified personnel participate in the TO Validation/Verification process when requested.

2.4.5.3. The 39th Information Operations Squadron (39 IOS) will:

2.4.5.3.1. Receive all new version updates of the CVA/H Weapon System and training developed by the PMO and 67 CW.

2.4.5.3.2. Evaluate each release to identify required changes to cyber training curricula as directed by ACC during periodic SRB, TPT, or as otherwise required.

2.4.5.3.3. Incorporate the latest CVA/H Type-1/difference training into the training curriculum IAW the CVA/H STP to reduce additional training requirements on operational units for members following the graduation of IQT.

2.4.5.4. Type-1 Initial Cadre will:

2.4.5.4.1. Actively participate, when available, in IPTs to ensure the training being developed is appropriate and adequate for the weapon system/crew position.

2.4.5.4.2. Provide timely responses to the PMO and/or vendor for pre-course verifications (e.g., provide information as requested so vendor can establish accounts if applicable) or technical tests (e.g., ensure students can access the content or venue if provided via distance learning).

2.4.5.4.3. Make every effort to attend training on the dates selected. Any course date changes or changes to Type-1 initial cadre must first be processed through the 67 CW/A3T and ACC/A3/2/6K. Do not go directly to the PMO or the vendor.

2.4.5.4.4. Upon completion of Type-1 training, provide feedback via the course survey (100% response).

2.4.5.4.5. Work with 67 CW/A3T and 39 IOS to update the MTTLs and develop training to incorporate new capabilities into MQT.

2.4.6. CVA/H Type-1 Training Process.

2.4.6.1. ACC/A3/2/6K, 67 CW/A3T , and operational squadron designated subject matter experts (SMEs) will participate in the acquisition process with the PMO and vendor.

2.4.6.1.1. 67 CW/A3T will select individuals for IPT participation via communications with squadrons.

2.4.6.1.2. Ensure all aspects of CVA/H weapon systems are considered during the requirements process (e.g., what other units will utilize/operate this capability and potentially need training).

2.4.6.1.3. Identify individuals who are competent and proficient to go through the full lifecycle of the training (e.g., request for proposal (RFP) package reviews, source selection, IPT participation, training requirements development, Type-1 training attendance, TO review, follow-on training development).

2.4.6.2. The PMO will notify ACC/A3/2/6K and coordinate on Type-1 Training to include dates, times, locations, and recommended prerequisites.

2.4.6.3. ACC/A3/2/6K, ICW, the 67 CW/A3T , will determine which weapon system operators require Type-1 training. To facilitate this, ACC/A3/2/6K will draft memos outlining prerequisites for potential attendees (Type-1 initial cadre) as well as responsibilities (e.g., development of follow-on training), and send out Task Management Tool (TMT) taskers that include the initial feedback survey (provided by PMO) to request attendees.

2.4.6.4. 67 CW/A3T will vet the unit-nominated Type-1 initial cadre to attend training and provide names to ACC/A3/2/6K via TMT; ACC/A3/2/6K will email the names to the PMO. Attendees must have the required experience, training, and qualifications to participate and provide feedback on the Type-1 training.

2.4.6.5. The PMO will confirm the logistical requirements.

2.4.6.6. Type-1 Initial Cadre will attend Type-1 training and produce required feedback to the PMO, ACC/A3/2/6K, and 67 CW.

2.4.6.7. The 67 CW and Type-1 initial cadre will review preliminary TO before publication (a 65% completion of draft TO is minimum via product (MVP) for publication) immediately following the Type-1 Training.

2.4.6.8. ACC/A3/2/6K will verify training completion and will draft Type-1 Initial Cadre Letters documenting training received, who received the training, and reiterating the responsibilities of Type-1 initial cadre to conduct difference training sufficient to qualify and certify the rest of the weapon system operators.

2.4.6.9. Training information will be incorporated into the necessary courses (IQT, MQT, etc.), as appropriate, IAW the CVA/H STP.

2.5. Senior Officer Qualification Training.

2.5.1. A Senior Officer is defined as a Deputy Group Commander or higher and must meet course entry pre-requisites as prescribed in **paragraph 2.1.1.1**. **(T-2)**

2.5.2. A Senior Officer in training, at the FTU or in-unit, is in a formal training status for the duration of the course(s). All non-training duties will be reassigned appropriately until training is completed. **(T-2)**

2.6. Squadron Leadership Training. Squadron Leadership includes the SQ/CC, Senior Enlisted Leader (SEL), and DO. Members assuming one of these positions at a unit that employs CVA/H will complete the following:

2.6.1. Successful completion of all IQT requirements IAW **paragraph 2.1.1**.

2.6.2. Successful completion of MQT requirements IAW **paragraph 2.1.2**.

2.6.3. Successful completion of INIT QUAL and MSN Evaluations IAW ACCI 17-202V2 and ACCMAN 17-2CVAHV2.

2.7. Requalification Training (RQT). RQT is any academic or positional training required to requalify for FMQ status. IAW ACCI 17-202V1, a cybercrew member is considered unqualified upon expiration of the member's required periodic evaluation or loss of currency exceeding six months, whichever occurs first. The duration of unqualified time is measured from when the cybercrew member became unqualified until the retraining start date. An unqualified cybercrew member can requalify IAW the following:

2.7.1. Unqualified up to 6 months: Completion of training in all delinquent items (as applicable), additional training as directed by the certifying official, and a Requalification Evaluation (RQ Eval) IAW ACCI 17-202V2 and ACCMAN 17-2CVAHV2.

2.7.2. Unqualified exceeding six months: Recompletion of MQT or appropriate RQT and an RQ Eval IAW ACCI 17-202V2 and ACCMAN 17-2CVAHV2. A SQ/CC may also require the completion of some or all portions of IQT before starting MQT or RQT if deemed necessary. **(T-2)** Applicable portions of IQT/MQT may be used to create a requalification program for cybercrew members who have been downgraded from FMQ or BMC status to address the deficiencies that caused the downgrade specifically.

2.8. CVA/H Cybercrew Operational Status. A cybercrew member may be assigned FT, FMQ, BMC, Non-Fully Mission Qualified (N-FMQ) or Non-Basic Mission Capable (N-BMC) status.

2.8.1. FT. A cybercrew member who completed their initial training being sent to their first operational unit. FT crewmembers will only perform RCP-tasked events or missions with instructor supervision. FT requirements include:

2.8.1.1. Successful completion of all IQT requirements IAW **paragraph 2.1.1**.

2.8.1.2. Individual currency (as applicable) IAW **paragraph 3.2**.

2.8.2. FMQ. An FMQ cybercrew member is available and qualified for the Squadron's mission. Cybercrew members can be assigned FMQ-Basic, Senior, or Master, depending on the proficiency level reached for their work role. Members achieving FMQ status are authorized to operate CVA/H unsupervised; however, FMQ-Basic members will remain in a supervised capacity by a Senior or Master-level qualified crewmember in the same work role while on a mission, per USCYBERCOM until Senior proficiency is earned. FMQ requirements include:

2.8.2.1. Successful completion of FT requirements IAW **paragraph 2.8.1**.

2.8.2.2. Successful completion of MQT requirements IAW **paragraph 2.1.2**.

2.8.2.3. Completed and certified USCYBERCOM Job Qualification Record (JQR) for target work role at Basic, Senior, or Master proficiency level.

2.8.2.4. Successful completion of INIT QUAL and MSN Evaluations IAW ACCI 17-202V2 and ACCMAN 17-2CVAHV2.

2.8.2.5. Individual currency (as applicable) IAW **Chapter 3** and the RTM.

2.8.2.6. FMQ-level proficiency standards IAW **Chapter 3** and the RTM.

2.8.3. BMC. A BMC cybercrew member is typically assigned to a MAJCOM, NAF, FTU, group/wing-level functions, squadron leadership positions, direct reporting unit, and supporting unit that employs the CVA/H weapon system. BMC status is meant to ease continuation training requirements for individuals needing to maintain a CVA/H operational status that does not conduct operations as their primary organizational mission. BMC requirements include:

2.8.3.1. Successful completion of FT requirements IAW **paragraph 2.8.1**.

2.8.3.2. Successful completion of MQT requirements IAW **paragraph 2.1.2**.

2.8.3.3. Completed and certified USCYBERCOM JQR for target work role at Basic, Senior, or Master proficiency level.

2.8.3.4. Successful completion of INIT QUAL and MSN Evaluations IAW ACCI 17-202V2 and ACCMAN 17-2CVAHV2.

2.8.3.5. Individual currency (as applicable) IAW **Chapter 3** and the RTM.

2.8.3.6. BMC-level proficiency standards IAW **Chapter 3** and the RTM.

2.8.3.7. CVA/H cybercrew members in BMC status must be able to attain FMQ status within 30 days (90 days for ARC). **(T-3)** The requirements to reach FMQ and BMC are the same except for the proficiency standards. BMC members will need to conduct appropriate RCP events to attain FMQ status.

2.8.4. N-FMQ / N-BMC. A cybercrew member that has been downgraded from FMQ/BMC status and requires additional training IAW **paragraph 3.2.3.2**. Cybercrew members in a N-FMQ/N-BMC status will only operate under instructor supervision. **(T-2)**

2.8.5. The decision to suspend, retain, or downgrade a cybercrew member's status for failure to meet standards established by this manual, ACCMAN 17-2CVAHV2, or ACCI 17-202V1 and V2 will be documented by citing all standards that apply. **(T-2)**

2.8.6. Certification by the certifying official is required to obtain FMQ/BMC statuses. Upon certification, the cybercrew member maintains FMQ/BMC status based on CT requirements IAW **Chapter 3**, the RTM, and ACCMAN 17-2CVAHV2.

Chapter 3

CONTINUATION TRAINING

3.1. Continuation Training (CT). CT consists of two aspects. The first provides cybercrew members with regular repetitions of the basic operational skills necessary to ensure the safe operation of the weapon system (i.e., currency). The second consists of regular repetitions of specific mission-related events required to accomplish the unit's assigned mission (i.e., proficiency). This manual and the current CVA/H RTM establish the minimum volume, frequency, and mix of cybercrew training requirements to maintain currency and proficiency in the assigned crew position and at designated certification/qualification levels. The SQ/CC will ensure each cybercrew member receives sufficient training to maintain individual currency and proficiency.

3.1.1. RTM. RCP is the formal CT program that provides the baseline for SQs to develop a realistic training program to meet all Designed Operational Capability (DOC) statement task requirements and specific NAF mission prioritization. The RTM is a formal document outlining the FY's RCP CT program. The RTM takes precedence over this volume and may contain updated requirements, missions, events, or tasks yet to be incorporated into this manual. The RTM applies to all CVA/H crewmembers. **(T-2)**

3.1.2. CT Requirements. Individual crewmembers are responsible for ensuring availability and preparedness for CT and for ensuring accuracy in tracking individual CT events. Cybercrew members should actively work with their supervisors, unit schedulers, and training offices to ensure the accomplishment of CT requirements.

3.1.3. Cybercrew members are responsible for reporting all accomplished training events to their attached unit. **(T-3)**

3.2. Currency. The first component of CT, maintaining currency, involves regular repetitions of the basic operational skills necessary to ensure the safe operation of the CVA/H weapon system. Currency tables for both Host and Network Analysts are in the annual RTM.

3.2.1. Currency may be established or updated by:

3.2.1.1. Accomplishing an event as a qualified cybercrew member whose currency has not expired.

3.2.1.2. Accomplishing an event under instructor supervision.

3.2.1.3. Events satisfactorily performed on any evaluation may be used to establish or update currency in that event.

3.2.1.4. All relevant operational activity should be counted and credited toward a cybercrew member's currency regardless of location (e.g., home station, exercise, contingency operations, training simulator).

3.2.2. Multi-position certified individuals must meet currency requirements for each cybercrew position to maintain operational statuses for assigned positions.

3.2.3. Recurrency. If a cybercrew member loses a particular currency, the member may not perform that event except to regain currency (see Recurrency Training in ACCI 17-202V1). Cybercrew members will accomplish overdue currency requirements before they are considered recertified to perform the task.

3.2.3.1. When cybercrew members become overdue on training requirements identified in the RTM, members will be downgraded to non-current for those events. If cybercrew members become non-current for events affecting FMQ/BMC status according to the RTM, regression to N-FMQ/N-BMC is required.

3.2.3.2. Members downgraded to a non-current status due to overdue training requirements must regain currency by performing the necessary training event under instructor supervision. **(T-3)**

3.3. Proficiency. The second component of CT, maintaining proficiency, consists of regular repetitions of specific mission-related events required to accomplish the unit's assigned mission. The RTM provides the baseline for SQ/CCs to develop a realistic training program to prioritize all DOC-tasked requirements to ensure team proficiency.

3.3.1. The RTM establishes the minimum proficiency requirements for a CVA/H cybercrew to maintain FMQ or BMC status. Standardized training event identifiers and descriptions are in Attachment 1 of the applicable RTM. Units may add unit-specific events as required. Descriptions of these events will be included in local training documentation. **(T-3)**

3.3.2. All operational activity will be counted and credited toward a cybercrew member's proficiency if the activity meets the RTM definitions for "effective," regardless of the location (e.g., home station, exercise, contingency operations, training simulator).

3.3.3. Individuals qualified in multiple crew positions will only receive credit for RCP events that align with the specific duties performed by the individual (e.g., a dual-qualified host and network analyst would satisfy host analyst proficiency requirements only for an event during which the member performed the duties of a host analyst).

3.3.3.1. End of Cycle Requirements. Cybercrew members who fail to complete event requirements by the end of the training cycle may require additional training depending on the type and magnitude of the deficiency. Refer to **paragraph 3.4** for proration guidance. In all cases, shortfalls will be reported to the Operations Group Commander (OG/CC). Failure to meet specific proficiency requirements will result in one of the following:

3.3.3.1.1. The SQ/CC or their designated representative will determine if the deficiency is significant enough to warrant downgrade to N-FMQ or N-BMC. FMQ status may be regained by satisfactorily performing all proficiency requirements under instructor supervision. These missions may also count toward the total requirements for the new training cycle.

3.3.3.1.2. Continuation of FMQ status may be authorized by the SQ/CC or their designated representative provided the member only accomplishes operational tasks that do not align to the incomplete event requirements. Training shortfalls will be noted in End-of-Cycle Reports, and cybercrew members will satisfactorily complete all training requirements as operations tempo allows and before performing any operational tasks that align to incomplete events.

3.4. Proration of Training.

3.4.1. Proration of End-of-Cycle Requirements. At the end of the training cycle, the SQ/CC may prorate any training requirements precluded by the following events: initial arrival date in Squadron, non-chargeable leave periods (e.g., parental, emergency, convalescent, etc.), non-

mission Temporary Duty (TDY), Permanent Change of Station (PCS), or non-mission exercises or deployments. Ordinary annual leave will not be considered as non-availability. Other extenuating circumstances, as determined by the SQ/CC, that prevent the crewmember from mission duties for more than 15 consecutive days may be considered non-availability for proration purposes. **(T-3)** The following guidelines apply:

3.4.1.1. Proration will not be used to mask training or planning deficiencies.

3.4.1.2. Proration does not apply to the ARC unless mobilized in support of an operational mission.

3.4.1.3. Proration is based on cumulative days of non-availability for mission duties in the training cycle and does not apply to individuals available for 15 days or less. Use **Table 3.1** to determine the number of months to be prorated based on each period of cumulative non-mission duty calendar days IAW ACCI 17-202V1. **(T-3)**

3.4.1.4. If MQT is re-accomplished, a crewmember's training cycle will start over at a prorated share following completion of a successful evaluation. **(T-3)**

3.4.1.5. No requirement may be prorated below one month. Prorated numbers resulting in fractions of less than 0.5 of a month will be rounded to the next lower whole number (one or greater). **(T-3)**

3.4.1.6. Newly assigned cybercrew members achieving FMQ/BMC status after the 15th of the month are in CT on the first day of the following month for proration purposes. A prorated share of RCP events must be completed. **(T-3)**

3.4.1.7. A cybercrew member's last month on station prior to a PCS may be prorated, provided one month's proration is not exceeded. PCS-ing individuals may be considered FMQ for reporting purposes during a period of 60 days from the date of the last training event or until loss of FMQ qualification, sign-in at the new duty station, or whichever occurs first. **(T-3)**

Table 3.1. Proration Allowance.

Cumulative Days of Non-Mission Activity	Proration Allowed (Months)
0 – 15	0
16 – 45	1
46 – 75	2
76 – 105	3
106 – 135	4
136 – 165	5
166 – 195	6
196 – 225	7
226 – 255	8
256 – 285	9
286 – 315	10
316 – 345	11
Over 345	12
Note: As an example, Capt Blix was granted 17 days of emergency leave in January and attended SOS in residence from March through April for 56 consecutive calendar days. The SQ/CC authorized a total of two months of proration from his training cycle (two months for the 73 cumulative days of non-availability).	

3.5. Additional Training. Any training that the Stan/Eval Examiner recommends to remedy deficiencies identified during an evaluation. Additional Training will be documented IAW **Chapter 5** and ACCMAN 17-2CVAHV2.

3.6. Currency and Proficiency Reporting. Currency and Proficiency are reported (IAW the RTM) through the Mid-Cycle and End-of-Cycle Reports, as directed by ACC/A3/2/6K. RTM Reporting instructions will take precedence over this volume.

3.6.1. Each cybercrew member will be reported as current or not current IAW currency tables in the RTM. All currency requirements must be up to date for the cybercrew member to count as current. The number of cybercrew members who are not current is then divided by the total number of cybercrew members.

3.6.2. Each cybercrew member will be reported as proficient or not proficient per event type (i.e., Hunt, Clear, Enable Hardening, and Assess) IAW the current RTM. All proficiency requirements must be up to date for the cybercrew member to count as proficient. The number

of cybercrew members who are not proficient is divided by the total number of cybercrew members.

3.6.3. Example: In a CPT with 27 qualified cybercrew members, one cybercrew member has not configured a *NIX firewall within the currency timelines. This cybercrew member is not current. Divide one by 27 (1÷ 27 = 0.037) and then multiply by 100 (0.037 x 100 = 3.7%). Record this percentage on the report. The unit will then perform the same calculation for proficiency requirements per event type.

Chapter 4

INSTRUCTOR TRAINING AND QUALIFICATION

4.1. Requirements. A qualified CVA/H instructor should be a competent SME in instruction methodology. The instructor shall be proficient in evaluating, diagnosing, and critiquing student performance, identifying learning objectives and difficulties, and prescribing and conducting remedial instruction. The instructor must be able to perform instruction across all training venues (e.g., classroom, training devices, operations (ops) floor, mission execution). Instructor trainees will be observed and supervised by a qualified instructor.

4.1.1. All instructors must be FMQ for at least 12 months prior to nomination (wing-level and below). **(T-3)**

4.1.2. Instructors must be current and qualified in any event they instruct. **(T-2)**

4.1.3. FTU instructors, including USAF Weapons School instructors, are only required to attain and maintain BMC. **(T-2)**

4.1.4. All instructors must adhere to USCYBERCOM Trainer requirements IAW USCYBERCOM CTM 7-0.1 *Joint Cyberspace Training and Certification Standards (JCT&CS)* and any subsequent guidance issued by USCYBERCOM or Air Force Cyber Command (AFCYBER).

4.1.5. IAW ACCI 17-202V1, units will develop an instructor upgrade syllabus approved by the OG/CC. It will minimally include:

4.1.5.1. Applicable equipment configurations and scheduling procedures (e.g., simulator, online equipment configuration, and instruction scenario control procedures). **(T-2)**

4.1.5.2. ISD processes and procedures. **(T-2)**

4.1.5.3. Requirements for trainees to develop, conduct, and administer classroom training, field training, simulator configuration and ops floors, as appropriate. **(T-2)**

4.1.5.4. Observance, at a minimum, of one certified instructor conducting academic training, in the simulator, on the ops floor, and in the field, as appropriate.

4.1.6. The SQ/CC will select instructor candidates based on experience, judgment, maturity, communication skills, operational skills, and technical knowledge of the weapon system.

4.2. Failure to Meet Instructor Requirements. Instructors may lose instructor status for the following:

4.2.1. Failure to instruct a course, additional training, or event at least once every 180 days.

4.2.2. The certifying official deems that loss of currency is of sufficient importance to require complete decertification (but not a complete loss of qualification).

4.2.2.1. If the affected cybercrew member retains instructor qualification IAW ACCI 17-202V2 and ACCMAN 17-2CVAHV2, recertification will be at the certifying official's discretion.

4.2.2.2. If an instructor loses currency, and the SQ/CC does not require decertification and removal from FMQ or BMC status, instructor status may be retained. The instructor will not instruct that mission or event until currency is regained.

4.2.3. Instructor's lack of ability. Instructors serve solely at the discretion of the SQ/CC. Instructors will exemplify a high level of performance and present themselves as reliable and authoritative experts. Instructors exhibiting substandard performance should be candidates for a suitability review for continued instructor duty.

4.2.3.1. Instructors will be decertified if the SQ/CC deems instructor is substandard, ineffective, or providing incorrect procedures or policy guidance.

4.2.3.2. Recertification requirements will be at the discretion and direction of the certifying official.

Chapter 5

DOCUMENTATION

5.1. Requirements. In all instances, using electronic forms, digital signatures, and electronic format Individual Training Folders (ITFs) is authorized.

5.1.1. Each block of training will be documented and maintained in the cybercrew member's ITF.

5.1.2. A cybercrew member's CT and additional training events are maintained in the member's ITF. Electronic format ITFs are authorized provided proper security measures, backup capability, and sustainment plans are in place.

5.1.3. Dispose of ITFs and other related material according to the RDS and AF guidance concerning the protection of PII.

5.2. Helpful Forms.

5.2.1. ACC Form 4419, *Record of Training*. Cybercrew member training events/tasks can be documented on the ACC Form 4419.

5.2.2. ACC Form 4420, *Individual's Record of Duties and Qualifications*. The ACC Form 4420 is an index providing pertinent training information extracted from all the ACC Forms 4419 accomplished by the cybercrew member.

DAVID G. SHOEMAKER, Maj Gen, USAF
Director of Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

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Cyber Vulnerability Assessment/Hunter (CVA/H) Ready Cybercrew Program (RCP) Tasking Memorandum (RTM), Fiscal Year (FY) 2024, 01 October 2023

Cyberspace Vulnerability Assessment/Hunter Weapon System Configuration Management Plan, Annex version 2.2., February 2018

Cyberspace Vulnerability Assessment/Hunter (CVA/H) System Training Plan (STP), 2024

Cyber Warfare Publication 3-33.4, *Cyber Protection Team Organization, Functions, and Employment*, 28 January 2020 (*Incorporating Change 1*, 18 May 2022)

DAFMAN 90-161, *Publishing Processes and Procedures*, 18 October 2023

DoD 5400.11-R, *Department of Defense Privacy Program*, 14 May 2007

Privacy Act of 1974 (5 USC § 552a, *Records Maintained on Individuals*), 31 December 1974

Prescribed Forms

None

Adopted Forms

ACC Form 4418, *Certificate of Cybercrew Qualification*

ACC Form 4419, *Record of Training*

ACC Form 4420, *Individual's Record of Duties and Qualifications*

DAF Form 8, *Certificate of Aircrew Qualification*

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

ACC—Air Combat Command

ACCI—Air Combat Command Instruction

ACCMAN—Air Combat Command Manual

AFCYBER—Air Force Cyber Command

AF—Air Force

AFI—Air Force Instruction

AFLCMC—Air Force Life Cycle Management Center

AFPD—Air Force Policy Directive

AFR—Air Force Reserve

AFRC—Air Force Reserve Command

AFRIMS—Air Force Records Information Management System

AFTTP—Air Force Tactics, Techniques, and Procedures

ANG—Air National Guard

ARC—Air Reserve Component

BCQ—Basic Cyber Qualified

BMC—Basic Mission Capable

CC—Commander

CL—Crew Lead

CMF—Cyber Mission Force

CMR—Combat Mission Ready

CNMF—Cyber National Mission Force

COTS—Commercial-Off-The-Shelf

CPT—Cyber Protection Team

CT—Continuation Training

CTA—Exploitation Threat Analyst

CTM—Cyber Training Manual

CV—Deputy Commander

CVA/H—Cyber Vulnerability Assessment/Hunter

CWP—Cyber Warfare Publication

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DAFMAN—Department of the Air Force Manual

DIP—Deployable Interceptor Platform

DOC—Designed Operational Capability

DO—Director of Operations

DoD—Department of Defense

ETCA—Education and Training Course Announcement

Eval—Evaluation

FMQ—Fully Mission Qualified

FT—Fully Trained

FTU—Formal Training Unit

FY—Fiscal Year

FYDP—Future Years Defense Program

GIP—Garrison Interceptor Platform

GP—Group Commander

HA—Host Analyst

HHQ—Higher Headquarters

HQ—Headquarters

IAW—In Accordance With

ICW—In Coordination With

INIT—Initial

IOC—Indicator of Compromise

IPT—Initial Planning Teams

IQT—Initial Qualification Training

ISD—Instructional Systems Development

IST—Initial Skills Training

ITF—Individual Training Folder

JCTM—Joint Cyber Tactics Manual

JQR—Job Qualification Record

JQS—Job Qualification Standard

LINUX—Linus Torvald's UNIX®

MAJCOM—Major Command

MDS—Mission Design Series

MFM—MAJCOM Functional Manager

MIP—Mobile Interceptor Platform

MOA—Memorandum of Agreement

MOU—Memorandum of Understanding

MQT—Mission Qualification Training

MSN—Mission

MTTL—Master Training Task List

MVP—Minimum Via Product

NA—Network Analyst

NAF—Numbered Air Force

N-BMC—Non-Basic Mission Capable

N-FMQ—Non-Fully Mission Qualified

NGB—National Guard Bureau

NIX—UNIX/LINUX

NLT—No Later Than

OG—Operations Group

OPCON—Operational Control

OPORD—Operations Order

OPR—Office of Primary Responsibility

ops—Operations

OSS—Operations Support Squadron

OTD—Operations Training Development

OTP—Operations Training Plan

PBED—Planning, Briefing, Execution, and Debriefing

PCAP—Packet Capture

PCS—Permanent Change of Station

PEX—Patriot Excalibur

PII—Personally Identifiable Information

PMO—Program Management Office

POC—Point of Contact

QUAL—Qualification

RCP—Ready Cybercrew Program

RDS—Records Disposition Schedule

RFP—Request for Proposal

RQT—Requalification Training

RTM—Ready Cybercrew Program Tasking Memorandum

SEL—Senior Enlisted Leader

SME—Subject Matter Expert

SORN—System of Record Notice

SOS—Squadron Officers School

MTP—Special Mission Training Plan

SMTP—Special Mission Training Plan

SMQ—Special Mission Qualification

SMT—Special Mission Training

SQ—Squadron

SRB—Syllabus Review Board

Stan/Eval—Standardization and Evaluation

STP—System Training Plan

T&R—Training and Readiness

TDY—Temporary Duty

TMT—Task Management Tool

TO—Technical Orders

TPT—Training Planning Team

U.S.—United States

USAF—United States Air Force

USCYBERCOM—United States Cyber Command

WST—Weapon System Team

Office Symbols

67 CW/A3T—67 Cyber Wing Training Office

ACC/A3—ACC Director of Operations

ACC/A3/2/6K—ACC Information Warfare Division

ACC/A3TO—ACC Flight Operations Branch

CNMF J6/8/9—CMFF

Terms

Basic—Basic proficiency level. An individual is considered qualified to be assigned to a work role. Basic qualification requires completion of all Air Force feeder training and outlined formal training; completion of all applicable basic level tasks outlined in the JQS (i.e., JQR Tab 4); and completion of all mission-specific training required to execute tasks for an assigned duty position.

Basic Cyber Qualified (BCQ)—A cybercrew member who has satisfactorily completed IQT and individual currency, as applicable. BCQ has been changed to FT to align with USCYBERCOM terminology.

Basic Mission Capable (BMC)—A cybercrew member who has satisfactorily completed IQT and MQT but needs to conduct operations as their primary organizational mission. This status is usually assigned to crew members with staff functions directly supporting cyber operations (e.g., wing staff, OSS personnel). CVA/H cybercrew members in BMC status must be able to attain FMQ (i.e., CMR) status within 30 days (90 days for ARC).

Certification—Procedure used to document competency in a particular task as determined by a certifying official. Not interchangeable with "qualification", which requires ACC Form 4418 or AF Form 8 *Certificate of Aircrew Qualification*, documentation.

Combat Mission Ready (CMR)—A cybercrew member who has satisfactorily completed IQT and MQT, and maintains certification, currency, and proficiency in the command or unit's operational mission. CMR has been changed to FMQ to align with USCYBERCOM terminology.

Continuation Training (CT)—Training that provides crew members with the volume, frequency, and mix of training necessary to maintain currency and proficiency in the assigned qualification level.

Currency—A measure of how frequently or recently a task is completed. Currency requirements should ensure the average cybercrew member maintains a minimum level of proficiency in each area.

Crew Lead (CL)—A CVA/H cybercrew member who is responsible for their crew and events. CL is an upgrade position and is a Host (HA) or Network Analyst (NA).

Cyber Threat Analyst (CTA) Special Mission Qualification—A CVA/H CTA tests and evaluates the security of network devices and host systems using advanced exploitation techniques and forensic analysis. A CTA's skills are focused on emulating threats to demonstrate vulnerability and risks present on a system or network. They use these same skills, to emulate the threat, to help identify adversary activity or attacks against host or networks systems. Like the HA they can create scripts but are focused on creating them for their test and evaluation purposes and across multiple types of systems in the environment.

Event—The leveraging of people, process, and technology resulting in the execution of cyber operations.

Fully Mission Qualified (FMQ)—A cybercrew member who has satisfactorily completed IQT and MQT, and maintains certification, currency, and proficiency in the command or unit's operational mission. Cybercrew members can be assigned FMQ-Basic, Senior, or Master, depending on proficiency level reached for their work role.

Fully Trained (FT)—A cybercrew member who has satisfactorily completed IQT and individual currency, as applicable.

Host Analyst—A CVA/H HA specializes in advanced vulnerabilities and threats in hardware and software of host systems. They have the skills and knowledge to assess the configuration, settings, and activity of a targeted host system to identify weak security, vulnerabilities, or maliciously configured systems. They have the expertise to utilize skills to investigate an incident or counter threat actions agnostic of operating system. They can assess a targeted host system for health and gather information which other operators can analyze or use to determine actions.

Master—Advanced proficiency level. Individual is fully trained and experienced in all aspects of the assigned unit mission. Individuals are required to complete the JQR to be qualified at the Master level proficiency. Certain work roles specifically note additional minimal requirements be met prior to qualifying as Master. Individual can supervise training and qualification of unit personnel on the execution of unit tactical and operational missions within their respective work roles. Master level proficiency includes advising leadership on mission challenges, direction, and risk mitigation strategies; development, oversight, and implementation of training to address technical competence shortfalls; evaluation of mission effectiveness; and, providing recommended solutions and implementing strategies to address gaps during mission execution.

Mission—The task, together with the purpose, that clearly indicates the actions taken and the reason, therefore. In common usage, a duty assigned to an individual or unit. The base mechanism used to achieve mission objectives is events. Missions may require multiple events from multiple units to accomplish the mission's objectives.

Network Analyst—A CVA/H NA specializes in advanced vulnerabilities and threats to network architectures, hardware, technologies, and associated systems. Their expertise allows them to reconstruct a network by analyzing the data collected from packet captures and flow data. They can identify misconfiguration & malicious activity based on the observable network traffic. They assess a network's configuration, settings, and activity to identify weak security, vulnerabilities, or maliciously configured network infrastructure. They are the experts to utilize for investigations into an incident or counter-threat actions on network infrastructure.

Proficiency—A measure of how well a task is completed. A cybercrew member is considered proficient when performing tasks at the minimum acceptable speed, accuracy, and safety levels.

Qualification Evaluation—Qualifies a cybercrew member to perform the duties of a particular crew position in the specified MDS. Requires ACC Form 4418 documentation.

Ready Cybercrew Program (RCP)—Annual event training requirements for cybercrew to maintain FMQ status.

Ready Cybercrew Program Tasking Memorandum (RTM)—A formal document that outlines the RCP Continuation Training (CT) program for the FY. This memorandum will be updated NTL two months before each new 12-month training cycle.

Senior—Intermediate proficiency level. To achieve Senior qualification, an individual must complete all outlined senior-level formal training and applicable senior-level tasks outlined in the JQS (JQR Tab 4) and complete all mission-specific training required to execute tasks for an assigned duty position. Individuals must also possess the experience and judgment to operate without direct technical oversight.

Specialized Mission Training—Training in any special skills (e.g., tactics, weapon system capabilities, responsibilities, etc.) necessary to carry out the unit's assigned missions that every crew member does not require. Specialized training is usually accomplished after the crew member is assigned FMQ or BMC status and is typically in addition to FMQ or BMC requirements. It may be an additional certification/qualification event as determined by the SQ/CC.

Status—Assigned to individuals based on the continuation training status (basic cyber qualification, basic mission capable, or mission ready), they are required to maintain.

Training Period—Any training period determined by the wing in which training requirements are performed.